

Java Programming

I&C SCI_X460.17

3.5 Units

Course Description

Java is a general-purpose computer-programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few implementation dependencies as possible. It is intended to let application developers "write once, run anywhere" (WORA), meaning that compiled Java code can run on all platforms that support Java without the need for recompilation. This course covers core language concepts including fundamental data types, flow control, standard function libraries, class and object design, inheritance, polymorphism, error handling, and file I/O. Object-oriented programming and modular design will be emphasized to support distributed development environments. Students will gain knowledge and experience in writing and designing programs using Java through hands-on exercises.

Recommended Knowledge

To be successful in this course, it is recommended that you possess the basic knowledge of programming.

Student Learning Outcomes

At the end of this course, students will be able to:

- Develop Java programs that are easy to debug and maintain
- Use an IDE to build Java programs using standard Java libraries
- Design object-oriented applications with Java
- Employ concepts of object-oriented programming such as encapsulation, inheritance, overriding, and polymorphism
- Manage error handling with exceptions
- Store and retrieve data with file handling
- Design graphical user interface
- Design programs that run with concurrency
- Design platform modules
- Use command line tools
- Debug Java programs

Course Materials

Required:

- Horstmann, C. (December 2021). *Core Java, Volume 1: Fundamentals*, 12th Ed. Oracle Press.
ISBN: 978-0137673629

- **Editor, Compiler, IDE**

To complete the assignments in this course students will need to use some form of code editing software, compiler, or an Integrated Development Environment (IDE), that students are free to choose their own.

IDEs also have additional features beyond code editing, and often have more robust editing features as well. Since they are more powerful tools though, they tend to also be more complex, and harder to learn.

[Eclipse IDE](#) is highly recommended to use for this course.

Course Outline

Module	Element	Description
Module 1	Key Topics	Review: Basics of OOP and Scanner class 1. Basics of OOP ○ Class/object, inheritance, polymorphism, abstraction , interfaces 2. Scanner class: object instantiation, data input (next(), nextLine(), nextBoolean(), nextInt(), nextDouble()), input verification (hasNext(), hasNextLine(), hasNextInt(), hasNextDouble()), examples 2. Scanner class ○ object instantiation, data input (next(), nextLine(), nextBoolean(), nextInt(), nextDouble()), input verification (hasNext(), hasNextLine(), hasNextInt(), hasNextDouble()), examples.
	Student Learning Outcomes	By the end of this module, you will be able to: ● Understand the basics of OOP ● Apply uses of the scanner class
	Module Elements	● Reading ● Lecture ● Discussion ● Lab

Module	Element	Description
	Assignments	Lab #1
Module 2	Key Topics	Inner Classes and Exceptions - Inner Classes: - Static / no static inner classes - members classes - anonymous classes - Exceptions - Objectives , Syntax, Error Vs Exception, Checked Vs Unchecked , Try / Catch/ Finally, Throw / Throws
	Student Learning Outcomes	By the end of this module, you will be able to: - Apply uses of the scanner class - Apply objectives behind inner classes and exceptions while maintaining the class structure
	Module Elements	- Reading - Lecture - Discussion - Lab
	Assignments	Lab #2
Module 3	Key Topics	Collections - General Architecture - Definition & objectives - Implementations : Hash table, Array List, Tree and Linked list - Interfaces : Set, List, Map - Collections: ArrayList, List, Set, TreeMap, TreeSet
	Student Learning Outcomes	By the end of this module, you will be able to: Define objectives of general architecture
	Module Elements	- Reading - Lecture - Discussion - Lab
	Assignments	Lab #3
Module 4	Key Topics	Useful/Standard Interfaces

Module	Element	Description
		- Comparable / Comparator interfaces - Definition & objectives - compareTo() and compare() methods - Sorting (Collections.sort()) - Iterable/Iterator interfaces - Definition & objectives - iterator() method - Running through a collection using iterator (hasNext())
	Student Learning Outcomes	By the end of this module, you will be able to: Apply uses of Useful and Standard Interfaces
	Module Elements	- Reading - Lecture - Discussion - Lab
	Assignments	Lab #4
Module 5	Key Topics	Generics - Generic class - Definition, its objective, how to use it, generic interfaces proposed by java (Collection, List, ..) - Diamond operator - Erasure - Inheritance - Generic methods: Syntax and examples - Wildcards: definition, objective, syntax and examples
	Student Learning Outcomes	By the end of this module, you will be able to: Apply uses of generic methods
	Module Elements	- Reading - Lecture - Discussion - Lab
	Assignments	Lab #5
Module 6	Key Topics	Lambda Built-in Functional Interfaces Use The built-in interfaces: java.util.function

Module	Element	Description
		○ Predicate, Consumer, Function, and Supplier ○ Filter a collection by using lambda expressions 2. Stream API : java.util.stream ○ creation, intermediate operations, final operations, performance 3. Reflection API - Introspection: java.lang.reflect ○ definition, dynamic access to information, reflection API & ○ annotations, performance & security
	Student Learning Outcomes	By the end of this module, you will be able to: ● Describe Lambda built-in functions ● Use Lambda expressions and streams
	Module Elements	● Reading ● Lecture ● Discussion ● Lab
	Assignments	● Lab #6
Module 7	Key Topics	Java Concurrency - General Architecture: ○ Definition & objectives ○ Implementations : Hash table, Array List, Tree and Linked list ○ Interfaces : Set, List, Map - Collections: ArrayList, List, Set, TreeMap, TreeSet ○ List and ArrayList : create, add, remove, run through, ... ○ Set : create, add, remove, ... ○ TreeMap, TreeSet: create, add, remove, run through, ... - Comparable / Comparator interfaces: ○ Definition & objectives ○ compareTo() and compare() methods ○ Sorting (Collections.sort())
	Student Learning Outcomes	By the end of this module, you will be able to: ● Describe and apply some types of collections ● Create, add, remove, runthrough List and ArrayList ● Create, add, remove Set ● Create, add, remove, run through TreeMap, TreeSet
	Module Elements	● Reading ● Lecture

Module	Element	Description
		• Discussion • Lab
	Assignments	• Lab #7

Evaluation and Grading

This course will be graded using the following weighted percentages for each of the assignments in the course. Your official course grade will be recorded in the UCI-DCE student portal. Do not rely on the Canvas gradebook as a representation of your official grade. Students pursuing a certificate program must take courses for a letter grade.

Assignments	% of Grade
Labs & Assignments	70%
Final Project	30%
Total	100%

Grading Scale

This course uses the following grading scale.

Letter Grade	Percentage
A+	97% - 100%
A	93% - 96%
A-	90% - 92%
B+	87% - 89%
B	83% - 86%
B-	80% - 82%
C+	77% - 79%
C	73% - 76%
C-	70% - 72%
D+	67% - 69%
D	63% - 66%
D-	60% - 62%
F	59% or less

Assignment due dates and late submissions

Due dates for assignments are indicated on the assignment submission page in canvas. Assignments submitted late will be docked points appropriately. Submissions past one week after the assignment due date will receive zero points (unless approved prior to assignment due date). Assignment submissions will not be accepted after the course has ended (see course end date listed for the course in your account at ce.uci.edu).

Technical Requirements

Below are the basic technical requirements for all UC Irvine Division of Continuing Education courses.

Hardware

To participate in this course, you need a computer or device with reliable internet access. The device should be able to play videos on the screen and audio through headphones or speakers.

Software

For this course, you must have Microsoft Office, Google Docs, OpenOffice, or another compatible word-processing software. If additional software is required, your instructor will provide detailed access information.

Skill Requirements

You may be required to use a web camera, copy and paste functions, attach documents, or use upload features in the LMS. In addition, you may need to upload and view documents in PDF format. You can [download Adobe Acrobat Reader](#) to open PDF files. (Please note that Apple OS X includes the Preview application, which allows you to open PDF files.)

UC Irvine Policies

Academic Performance

Successful completion of an ACP program requires a passing grade of “C” (73%) or higher in every course. A student with a grade of “C-” (72%) or less in any course will not complete the program. This may negatively impact any planned activities the following quarter—e.g., second ACP, internship, OPT, etc.

Punctuality & Attendance

Instructors will mark student attendance and monitor participation for each class meeting. To be considered

“present,” students must actively participate in class activities and arrive prepared with required materials.

- **Punctuality:** Students are expected to attend class on time and remain until the end of class. Arriving late or leaving early may negatively impact your grade.
- **Absence:** Students are expected to attend class each day. Instructors will mark attendance daily. If you miss class, it may negatively impact your grade. Attending and actively participating in class is an important part of the ACP program.

Course Policies

Professionalism is expected during class. Professionalism includes conducting yourself in a mature manner. Students are expected to maintain respectful behavior during class. The following behaviors are not acceptable:

1. **Tardiness:** It is rude to interrupt instruction by walking in late. I expect that you will arrive to class on time. If you do arrive late, I expect you to take your seat quietly and not disrupt the class.
2. **Noise and Distractions:** Keep necessary conversation to a minimum and do not create any distractions, particularly during group presentations. Be courteous to those around you.
3. **Use of cell phones:** You may bring your cell phone to the class; however, it must be set to silent. Please do not answer phone calls or texts during the class unless there is an emergency. If you have an urgent situation that requires you to be available by phone, please excuse yourself from class before taking the call.
4. **Use of laptop:** If you bring your laptop to class, it is not acceptable to surf the net or send emails during the class.

Plagiarism

Plagiarism is intellectual theft and is not acceptable. It means the use of the intellectual creations of another without proper attribution. Plagiarism may take two main forms, which are clearly related:

1. To steal or pass off as one's own the ideas or words, images, or other creative works of another.
2. To use a creative production without crediting the source, even if only minimal information is available to identify it for citation.
3. Credit must be given for every direct quotation, for paraphrasing or summarizing a work (in whole, or in part, in one's own words), and for information which is not common knowledge.

Recording & Electronic Devices

No recording (audio, video, photography) is allowed in class. Unless approved by the instructor, all electronic devices (other than laptops/tablets used to access course materials) must be silenced and put away in a location that is out of sight.

Code of Conduct

All participants in the course are bound by the [University of California Code of Conduct](#).

Academic Honesty Policy

The University is an institution of learning, research, and scholarship predicated on the existence of an environment of honesty and integrity. As members of the academic community, faculty, students, and administrative officials share responsibility for maintaining this environment. It is essential that all members of the academic community subscribe to the ideal of academic honesty and integrity and accept individual responsibility for their work. Academic dishonesty is unacceptable and will not be tolerated at the University of California, Irvine. Cheating, forgery, dishonest conduct, plagiarism, and collusion in dishonest activities erode the University's educational, research, and social roles.

Students who knowingly or intentionally conduct or help another student engage in dishonest conduct, acts of cheating, or plagiarism will be subject to disciplinary action at the discretion of UCI Division of Continuing Education.

AI Use Statement

DCE supports the ethical and appropriate use of AI when it enhances learning, strengthens professional skills, and aligns with course learning objectives. To learn more about AI use in your course, please review the [AI Learner Use Statement](#).

Disability Services

At the University of California, Irvine, providing a culture of inclusion and equal opportunity for students with disabilities is a campus-wide responsibility and commitment. The Disability Services Center (DSC) provides reasonable accommodations and connects students with other valuable campus resources. Accommodations are designed to remove barriers and ensure access to programs for all students with permanent or temporary disabilities.

A UCInetID is required for students to begin the intake process. Contact Mike Sciarappa in the DCE Student Affairs & Student Services Office at healthandwellness@ce.uci.edu or 949-824-6872 for assistance with obtaining a UCInetID and registering with the Disability Services Center. If you have any questions regarding accommodations, documentation, and resources, please contact the DSC as soon as possible at 949-824-7494. You can also visit our [Students with Disabilities webpage](#) for more information.

Privacy

UC Irvine is fully committed to maintaining the integrity of your personal student information, including academic records, in accordance with the Federal Family Education Rights and Privacy Act of 1974 (FERPA). For complete information on privacy and FERPA, please visit the [UCI Privacy and Student Records website](#).

Accessibility

The University of California and UC Irvine are committed to improving accessibility for all students. For complete information about UC Irvine's policy affecting all information technology systems and software, please visit [UCI's accessibility policy](#). For accessibility information pertaining to the software used in UCI courses, please see below:

- [Canvas' Accessibility Statement](#)
- [Zoom Video Communications Accessibility Conformance Report](#)
- [Yuja Accessibility Documents](#)

UCI-DCE Student Services

The UCI Division of Continuing Education Student Services office is responsible for maintaining enrollment, academic, and financial records while providing academic support services to DCE students, instructors, and staff. Services include:

- Student records and DCE My Account portal access
- Adding, dropping, and waiting lists for all DCE courses and programs
- Cashiering, refunds, and accounts receivable
- Grades, transcripts, candidacy, and certificates
- International F-1 student advising, student academic success, and wellness support
- Important notifications regarding DCE course-related changes such as schedule, location, instructor, and course materials

Descriptions of many of these services can be found on [UCI-DCE's Student Resources webpage](#). Additionally, UCI-DCE Student Services is available by email (dce-services@uci.edu).